1 Technical specification

1.1 General

Supply voltage Power consumption Supply frequency Ambient temperature range	120/230Vac +10% - 15% Approx. 60VA 50/60 Hz ±5% 0 to 60°C (32 to 140°F)
Control unit protection category	IP20. The control must be situated in a clean environment according to EN6730-1. Indoor: Control must be mounted in an NEMA1 (IP40) enclosure Outdoor: Control must be mounted in an NEMA3 (IP54) enclosure
Unit dimensions	Control unit 176 x 115 x 92mm (6.93 x 4.53 x 3.62 in) deep Display 132 x132 x 36mm (5.20 x 5.20 x 1.42 in) deep
Weight	Control unit 1.55kg (3.42 lbs) Display 0.90Kg (1.98 lbs)
Type of display	2 lines x 20 characters, vacuum fluorescent display with membrane keypad.

1.2 Servo-motor control – IMPORTANT: See Section 3.2

Type	CANbus
Maximum (total) number of servo-motors	10 (see section 3 regarding servo-motors)
Maximum number of fuel profiles	4
Max. number of setpoints per profile	24 including close, purge and ignition
Positioning accuracy	+/-0.1°
Response time to positioning error	15s for ±1.0°, 1s for ±5.0°
Number of fuel motors	Not Limited by profile (e.g. 2 or 3 fuel motors). All non-monotonic.
Number of air (non fuel) motors	Not Limited by profile (e.g. 2 or 3 fuel motors). All non-monotonic. **NOTE: Number of servomotors is limited in all profiles by VA rating. See Section 3.1**

Interface to PPC6000	CANbus
Speed	30 seconds for 90 degrees.
Microswitches	Open & close positions
Torque	(see torque table below)

	Driving	Holding
NXC04	4Nm/ 3 ft/lb	2Nm
NXC12	12Nm/ 9 ft/lb	18Nm
NXC20	20Nm/ 14.7 ft/lb	18Nm
NXC40	40Nm/ 29 ft/lb	20Nm

Voltage:	24-30Vac supplied from control only
<u>VA rating</u>	NXC04 = 3VA NXC12 = 5VA NXC20 = 10VA NXC40 = 18VA
Protection Category	NXC04 = IP40, NEMA 1 NXC12, NXC20 = IP54, NEMA 3S NXC40 = IP65, NEMA 4
Typical accuracy	+/-0.1°
Accuracy (as specified by EN12067)	+/-0.5°

1.3 Digital outputs *(PPC6000)*

Controlled Shutdown, Safety Shutdown	
Type	On-off relay, de-energize for off.
Minimum current	200mA rms
Maximum current (per output)	8A rms
Maximum voltage	250Vac rms

1.4 Alarm output *(PPC6000)*

Alarm	
Type	On-off relay, de-energize for off.
Minimum current	200mA rms
Maximum current (per output)	4A rms
Maximum voltage	250Vac rms

1.5 Digital inputs *(PPC6000)*

Low Voltage digital inputs 1 to 4, HIGH input, AUTO input.	Digital, Switching 0V to 5V pulsed. Feed must be taken from the correct terminal as indicted in this manual. Inputs 1-4 configurable for 4-20Ma. Via function block programming. Less than 25mA ±10V absolute maximum
Profile Select High voltage inputs	Digital, 0V for off, 90-264Vac for on.

1.6 Pressure/Temperature Input (*PPC6000)*

0-5V Maximum current Maximum voltage Input accuracy (typical) Input accuracy (as specified by EN12067)	 Less than 2mA 0 to 5.0 volts maximum. ±0.1% ±0.3%
4-20mA Maximum current Burden (load) resistor Input accuracy (typical) Input accuracy (as specified by EN12067)	2 wire loop or ext. powered 0 to 25mA maximum 220 Ohm nominal ±1.0% ±1.1%

1.7 Communications interface *(PPC6000)*

2-wire RS485 plus ground, with termination resistor selected using a link. See Section 2.5.5.

An isolated 2-wire RS485 interface is available as an option. See Section 2.5.6.

See MOD-6101 (FIREYE serial communications protocol manual) for details.

1.8 Optional Oxygen Probe Interface Unit *(NXO2INT) - optional*

Supply voltage	115/230Vac ± 15%
Power consumption	Approximately 60VA
Supply frequency	50/60Hz ± 5%
Ambient temperature range	0 to 60°C (32 to 140°F)
Protection category	NEMA4 (IP65).
Unit dimensions	160 x 98 x 63mm (6.30 x 3.86 x 2.48") deep
Weight	1.34Kg (2.95 lbs)
Interface to PPC6000 series.	FIREYE specific CANbus.
Interface to oxygen probe.	FIREYE specific or 4-20mA

1.9 Optional Ambient Air Temperature Unit *(NXIATS) - optional*

Type	CANbus
Ambient temperature range	0 to 60°C (32 to 140°F)
Protection category	NEMA3 (IP54)
Unit dimensions	57 x 63 x 35 mm (2.25 x 2.5 x 1.37 inches)
Weight	0.15 kg (5.4 oz.)
Interface to PPC6000 series.	FIREYE specific CANbus.

1.10 Variable Speed Drive (VSD) Daughter Board *(NXDBVSD) - optional*

Ambient temperature range	0 to 60°C (32 to 140°F)
Protection category	Not applicable (fits inside PPC6000 unit).
Analog inputs (4 – 20mA)	3 max (non-isolated)
Input impedance	120ohms
Analog outputs (4 – 20mA)	3 max (isolated)
Maximum loop resistance	250 ohms
Isolation voltage	50v
RS485 communications.	Modbus RTU

1.11 Fireye NXC04, NXC12, NXC20, NXC40 Servo Motors

Interface to PPC6000	CANbus
Speed	30 seconds for 90 degrees.
Microswitches	Open & close positions
Torque	see table below
Voltage:	24-30Vac supplied from control only
VA rating	NXC04 = 3VA NXC12 = 5VA NXC20 = 10VA NXC40 = 18VA
Protection Category	NXC04 = IP40, NEMA 1 NXC12, NXC20 = IP54, NEMA 3S NXC40 = IP65, NEMA 4
Typical accuracy Accuracy (as specified by EN12067)	+/-0.1° +/-0.5°

	Driving	Holding
NXC04	4Nm/ 3 ft/lb	2Nm
NXC12	12Nm/ 9 ft/lb	18Nm
NXC20	20Nm/ 14.7 ft/lb	18Nm
NXC40	40Nm/ 29 ft/lb	20Nm

1.12 Fireye NXO2INT Oxygen Probe Interface Unit (optional)

Supply voltage	115/230Vac ± 15%
Power consumption	Approximately 60VA
Supply frequency	50/60Hz ± 5%
Ambient temperature range	0 to 60°C (32 to 140°F)
Protection category	NEMA4 (IP65).
Unit dimensions	160 x 98 x 63mm (6.30 x 3.86 x 2.48“) deep
Weight	1.34Kg (2.95 lbs)
Interface to PPC6000 / NX6100	FIREYE specific CANbus.
Interface to oxygen probe.	FIREYE specific or 4-20mA

1.13 Fireye NXIATS Ambient Air Temperature Sensor (optional)

Type	CANbus
Ambient temperature range	-29°C to 60°C (-20°F to 140°F) **NOTE:** Accuracy below 0°C (32°F) may vary slightly. Agency testing conducted to **0°C *only*.**
Protection category	NEMA3 (IP54)
Unit dimensions	
Weight	
Interface to PPC6000 / NX6100	FIREYE specific CANbus.

1.14 Approvals

Tested in accordance with the Gas Appliance Directive (GAD 90/396 EEC), encompassing the following standards:

- SIL Level 3 – Kiwa Gastec Report #123836
- ANSI/UL 462, Heat Reclaimers for Gas, Oil, or Solid Fuel-Fired Appliances
- ANSI/UL 1995, Heating and Cooling Equipment
- ANSI/UL 1998, Software in Programmable Components
- CAN/CSA-C22.2 No. 236, Heating and Cooling Equipment
- FM
- ENV1954, Internal and external behavior of safety related electronic parts
- EN60730-1, Automatic electrical controls for household and similar use
- prEN12067, Gas/air ratio controls for gas burners as gas burning appliances
- SIL level 3 per Kiwa report #123836

1.15 Parts List with Description

PART NO*	DESCRIPTION
PPC6000 FUEL AIR RATIO CONTROLLER	
PPC6000	Stand-alone parallel positioning controller, with up to ten (10) selectable function CANbus servo-motor outputs. Includes user configurable function blocks for custom applications. Display ordered separately.
DISPLAY MODULES FOR PPC6000	
NX610	CANbus display for PPC6000 with upload/download of PPC6000 data and three programmable relays
NXTSD104**	10.4" Touchscreen Display with upload/download, full commissioning, data log, internet connection, four programmable relays, 10 line voltage Digital Inputs.
SERVO-MOTORS FOR PPC6000	
NXC04	4 wire CANbus Servo-motor, 3 ft lbs. torque, 4 Nm, 50/60 Hz, 24 VAC.

PART NO*	DESCRIPTION
NXC12	4 wire CANbus Servo-motor, 9 ft lbs. torque, 12 Nm, 50/60 Hz, 24 VAC.
NXC20	4 wire CANbus Servo-motor, 14.75 ft lbs. torque, 20 Nm, 50/60 Hz, 24 VAC.
NXC40	4 wire CANbus Servo-motor, 29.5 ft lbs. torque, 40 Nm, 50/60 Hz, 24 VAC.
EXPANSION INTERFACE MODULES FOR PPC6000	
NXDBMB	Modbus RTU communications card
NXDBVSD	VSD interface daughter board with two VSD channels, one analog output, two counter inputs, two programmable relays, isolated RS485, Modbus RTU communications
NXO2INT	CANbus O2 interface module with Fireye and generic (4-20mA) probe inputs.
O2 PROBES FOR PPC6000	
NXO2PK4	O2 probe assembly (for flues 300mm to 1000mm). Includes NXIATS CANbus ambient temperature sensor, flange kit.
NXO2PK6	O2 probe assembly (for flues 600mm to 2000mm). Includes NXIATS CANbus ambient temperature sensor, flange kit.
NXO2PK8	O2 probe assembly (for flues 1200mm to 4000mm). Includes NXIATS CANbus ambient temperature sensor, flange kit.
NXIATS	PPC6000 CANbus Inlet (ambient) Air Temperature Sensor Sensor -29°C to 60°C (-20°F to 140°F)
SENSORS FOR PPC6000	
PXMS-15	Steam Pressure Sensor: 0 - 15 PSI, 0 - 1 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC6000).
PXMS-200	Steam Pressure Sensor: 0 - 200 PSI, 0 - 14 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC6000).
PXMS-300	Steam Pressure Sensor: 0 - 300 PSI, 0 - 21 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC6000).
BLPS-15	Steam Pressure Sensor: 0 - 15 PSI, 0 - 1 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC5000 / PPC6000 / NX3100 / NX4100 / NX6100).
BLPS-30	Steam Pressure Sensor: 0 - 30 PSI, 0 - 1 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC5000 / PPC6000 / NX3100 / NX4100 / NX6100).
BLPS-200	Steam Pressure Sensor: 0 - 200 PSI, 0 - 14 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC5000 / PPC6000 / NX3100 / NX4100 / NX6100).
BLPS-300	Steam Pressure Sensor: 0 - 300 PSI, 0 - 21 bar, 4-20mA output, 1/2" NPT, non self-check (for use with PPC5000 / PPC6000 / NX3100 / NX4100 / NX6100).
TS350 (-2), (-4), (-8)	Temperature Sensor, Range 32°F-350°F (0-176°C), 4-20mA linear output, includes 1/2 - 14 NPT well. See bulletin BLZPTS-1 for complete description.
TS752 (-2), (-4), (-8)	Temperature Sensor, Range 32°F-752°F (0-400°C), 4-20mA linear output, includes 1/2 - 14 NPT well. See bulletin BLZPTS-1 for complete description.
SOFTWARE	
NXAM	ComFire communications software on CD for Nexus and PPC controls.
COMMUNICATION INTERFACE GATEWAY	
NXDBVSD	VSD interface daughter board with two VSD channels, one analog output, two counter inputs, two programmable relays, isolated RS485 – Modbus RTU communications
NXMBIV2	Modbus RTU Communications daughter board.

* FOR ADDITIONAL PARTS SEE FIREYE PRICE BOOK CG-14

** SOME FEATURES INCLUDED ON NXTSD MANUFACTURED AFTER MAY 2011.

2 Installation

This section contains basic installation information concerning choice of control and servomotor environment, wiring specification and connection details.

WARNING

EXPLOSION OR FIRE HAZARD
CAN CAUSE PROPERTY DAMAGE,
SEVERE INJURY OR DEATH

To prevent possible hazardous burner operation, verification of safety requirements must be performed each time a control is installed on a burner, or the installation modified in any way.

This manual may cover more than one model in the PPC6000 controls. Check for Additional Information at the end of this chapter.

This control must not be directly connected to any part of a Safety Extra Low Voltage (SELV) circuit.

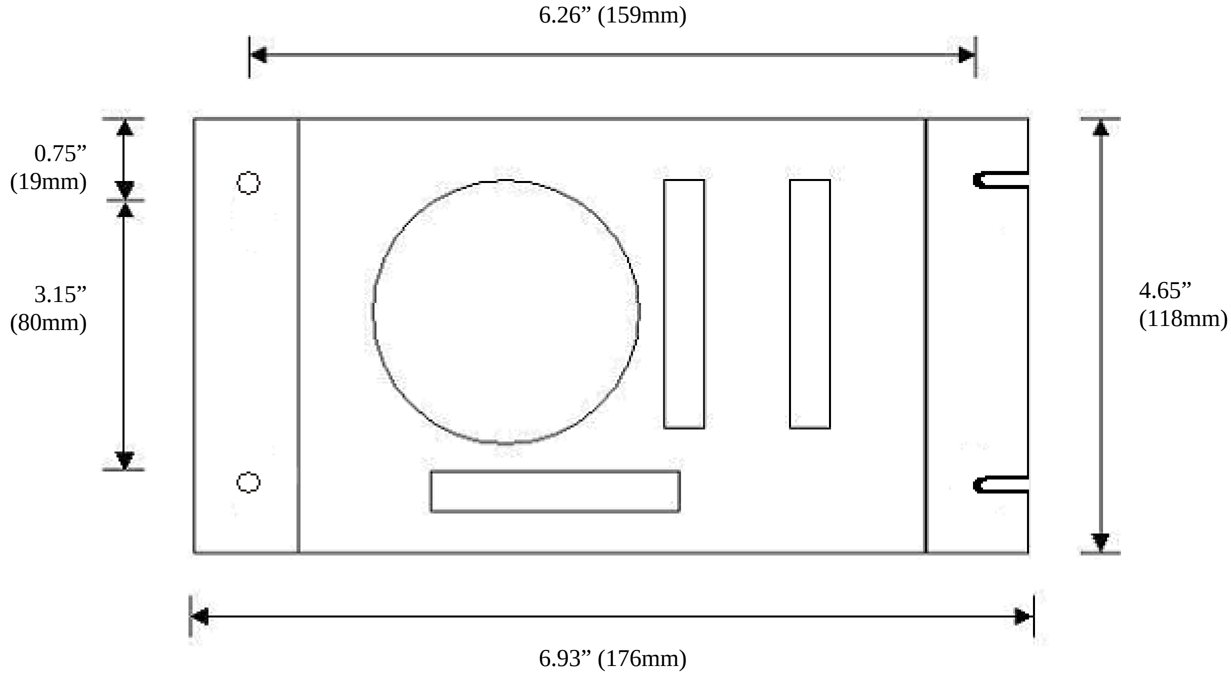
WHEN INSTALLING THIS PRODUCT:

- Read these instructions carefully and ensure you fully understand the product requirements. Failure to follow them could damage the product or cause a hazardous condition.
- Check the ratings given in these instructions to ensure the product is suitable for your application.
- After installation is complete, check the product operation is as described in these instructions

CAUTION

- Disconnect the power supply before beginning installation to prevent electrical shock, equipment and/or control damage. More than one power supply disconnect may be involved.
- Wiring must comply with all applicable codes, ordinances and regulations.
- Loads connected to the PPC6000 series must not exceed those listed in the specifications as given in this manual.
- All external components connected to the control must be approved for the specific purpose for which they are used.

2.1 Mounting details for the PPC6000 control

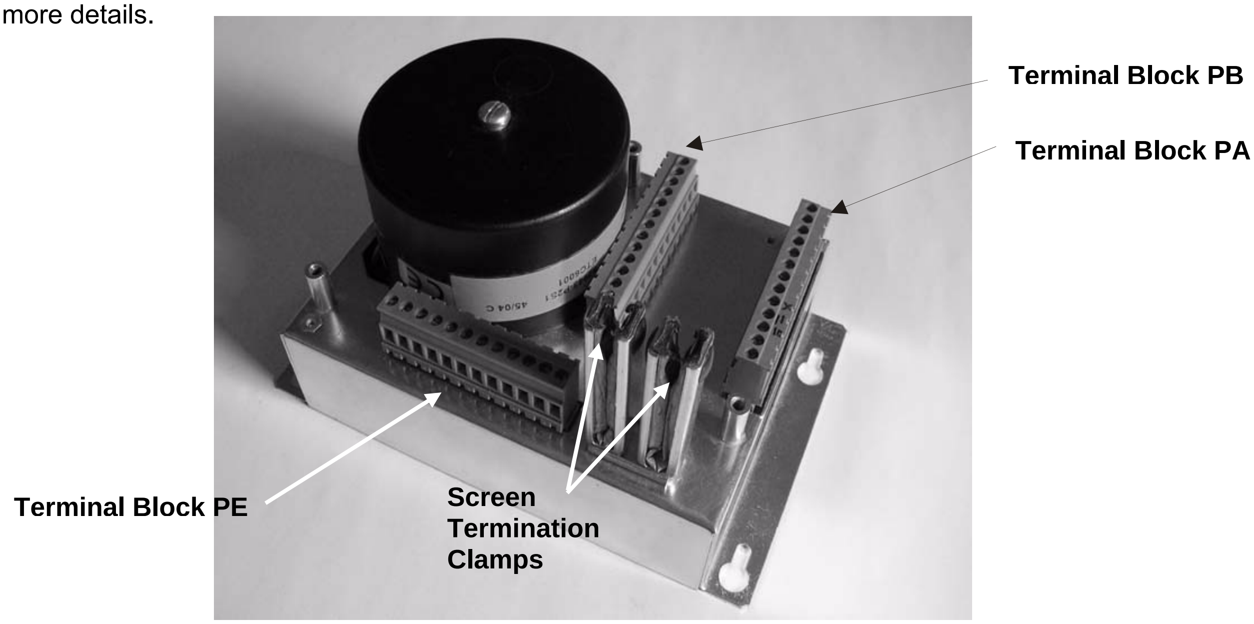


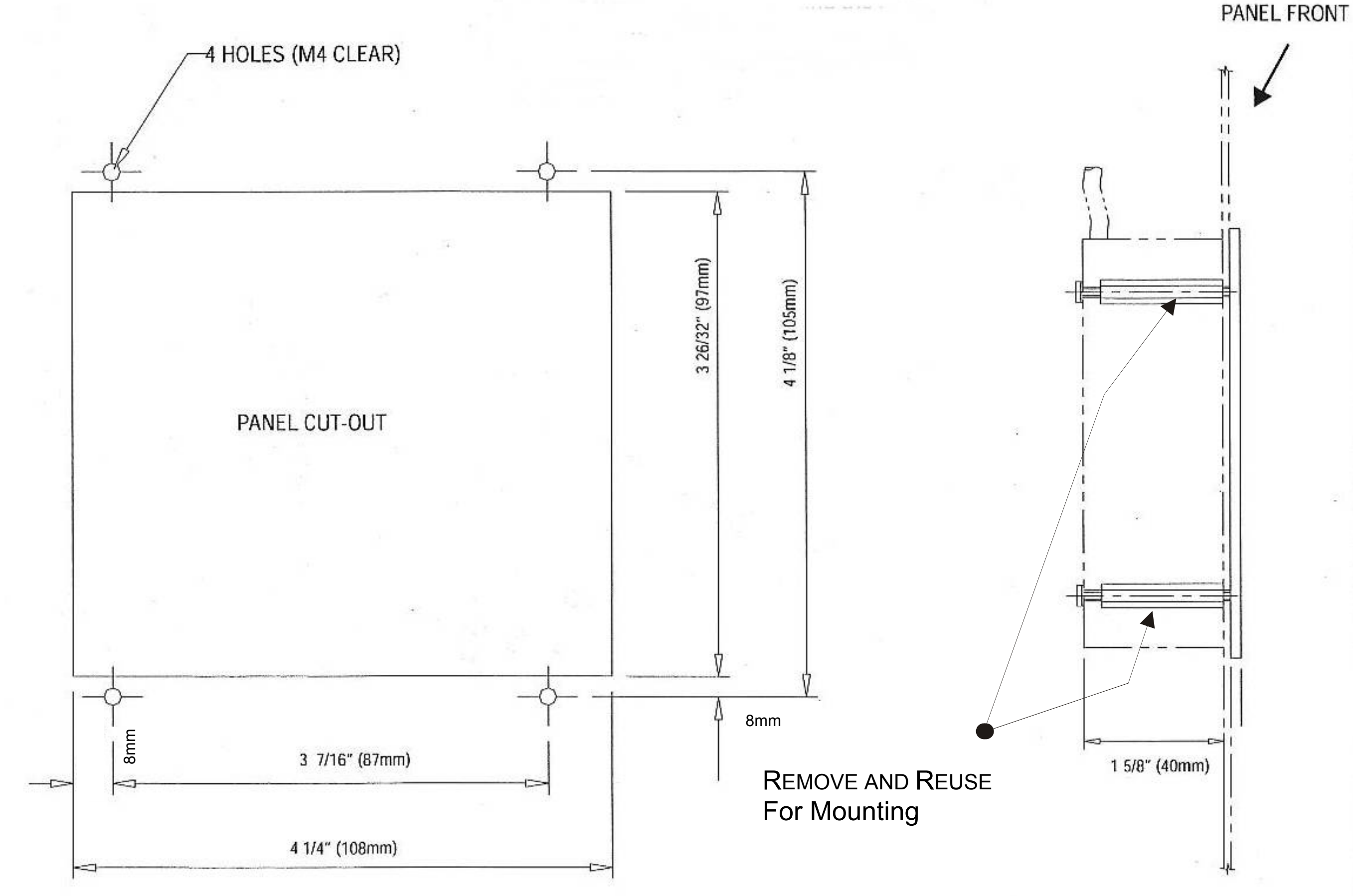
A Template for mounting is provided at the end of this manual for convenience. See Section 8.

There are two versions of the control, one that is intended to be mounted inside of a burner control cabinet, the other which has provision for conduit storage of field wiring etc. that can be mounted without the need for a burner control cabinet. If the version which is designed to be fitted inside of a burner control cabinet is being used the cabinet should have a minimum protection level of NEMA1 (IP40) for indoor use or NEMA3 (IP54) for outdoor use.

The control can be mounted in any attitude; clearances of a least 2.36-inch (60mm) should be left around the unit to allow sufficient space for wiring and to ensure reliable operation.

The ambient operating temperature range of the equipment is 0 to 60°C (32 to 140°F). Refer to section 7 for more details.





The display is held in using the four brass standoffs. The panel is cut out inside the four mounting holes only. Remove the four brass standoffs, insert the screw studs through the mounting holes, then re-install the brass standoffs. **Do not over tighten the standoffs.**

A Template for mounting is provided at the end of this manual for convenience. See Section 8.